Sohan Jayram Reddy Kadiri

Seattle, Washington, United States | sohanjreddy@outlook.com | (206) 380-6390 | github.com/sohanreddyk | linkedin.com/in/sohanreddyk | www.sohanreddy.org/

PROFESSIONAL SUMMARY

Software Engineer specializing in backend development and distributed systems, with experience building REST APIs, real-time services and scalable applications using Java, Go, Node.js and PostgreSQL. Focused on performance, reliability and fault tolerance.

EDUCATION

Northeastern University Sep 2025 - May 2027
Master of Science, *Computer Science* GPA: 3.9

Siddaganga Institute of Technology Dec 2021 - Jun 2025
Bachelor's, *Artificial Intelligence and Data Science*

WORK EXPERIENCE

Siemens | Software Engineer Intern | Remote Aug 2024 - Jul 2025

- Engineered backend services using **Node.js, REST APIs, and PostgreSQL** across 2 production applications, supporting 2,500+ users.
- Integrated a **Pinecone-backed RAG pipeline** into an AI chatbot, improving retrieval relevance by 40% and supporting 1,200+ conversations at 850 ms p95 latency.
- **Led 3 interns** through backend architecture and API integration, resolving 9 production issues through log analysis, debugging, and service-level testing.
- Automated deployments using **Docker, Azure, and Vercel CI/CD**, reducing release time by 70%, from 20 to 6 minutes.

Unisys | Software Engineering Intern | Remote Apr 2024 - Jun 2024

- Developed **RESTful backend services** for a digital marketplace serving 8,000+ users and processing 1,600+ orders, generating $20K in gross merchandise value.
- Optimized **checkout APIs and transaction workflows** through one-click checkout and funnel improvements, increasing conversion by 22% and contributing $14K in revenue.
- Engineered a **WebSocket-based messaging system** supporting 350 concurrent connections, sub-150 ms message latency, and 99.8% delivery reliability.
- Optimized **database queries and API response handling**, reducing p95 request latency by 45%, from 400 ms to 220 ms.

PROJECTS

Insight — Distributed Real-Time Analytics Platform | TypeScript, NestJS, Kafka, PostgreSQL, Redis, Docker

- Architected a **multi-tenant event-processing platform** using **NestJS, Kafka, and PostgreSQL**, with **REST APIs**, **Redis**-backed rate limiting, and real-time event delivery.
- Implemented **idempotent Kafka consumers** and **transactional PostgreSQL aggregation**, preventing duplicate event counts and routing failed messages to a dead-letter queue.
- Increased processing throughput by **61×**, from 46 to 2,816 events/sec, using Kafka batch consumption, bulk PostgreSQL writes, and pipelined Redis operations.

TaskForge — Distributed C++ Task Queue | C++17, Redis, Multithreading, Python

- Engineered a **Redis-backed task queue in C++17** with multithreaded workers, priority scheduling, asynchronous execution, and at-least-once delivery.
- Implemented **configurable retries and stuck-task recovery** to restore interrupted jobs, achieving 99.9% successful task completion during failure testing.
- Optimized **multithreaded worker scheduling and concurrent task execution**, sustaining 5,000 tasks/sec at 20 ms p95 dispatch latency.

Harbor — Distributed LLM API Gateway | Go, Redis, Python, FastAPI, Docker, Prometheus

- Engineered an **OpenAI-compatible API gateway in Go** with streaming, model routing and semantic caching, supporting 1,000 requests/min across 3 inference providers.
- Implemented **Redis-backed rate limiting, circuit breakers, and automatic failover**, successfully handling 400/400 requests during a simulated primary-provider outage.
- Accelerated cached responses by **9.5×** through embedding-based semantic caching, achieving an 85% cache-hit rate under a benchmark workload using a mock provider.

SKILLS

Languages: Java, Python, Go, C++, TypeScript, JavaScript, SQL
Backend & Frameworks: Node.js, NestJS, FastAPI, Spring Boot, REST APIs, WebSockets, Microservices
Databases & Messaging: PostgreSQL, Redis, MongoDB, Kafka, RabbitMQ
Cloud & Infrastructure: AWS (EC2, S3, Lambda), Azure, Docker, Kubernetes, CI/CD, GitHub Actions
Engineering: Distributed Systems, Multithreading, Concurrency, Caching, Rate Limiting, Fault Tolerance, Git

CERTIFICATIONS & ACHIEVEMENTS

- Open Source: Merged PR #472 into mindsdb/anton, fixing a bug that deleted saved user rules and adding 3 regression tests. Sep 2026
- Publication: "Detection of DDoS Attacks in Fog Computing using Interpretable Quantum-based 2Q Learning Approach" - IEEE Jan 2025